Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/11/2006
PI Name:	Behringer, Robert P. Ph.D.		
Project Title:	Gravity and Granular Materials		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27708-0305	**Congressional District:**	1
Comments:	NOTE: PI Robert Behringer is deceased, as of July 2018.		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	05/02/2004	**End Date:**	12/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Wilkinson, R Allen	**Contact Phone:**	216-433-2075
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Flight Program:			
Flight Assignment:	Note- grant renewed for 4 years/new NAG#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kondic, Lou (New Jersey Institute of Technology)		
Grant/Contract No.:	NNC04GB08G		
Performance Goal No.:			
Performance Goal Text:			
	The initial task consisted of a detailed characterization of the dynamics of shaken granular materials. Experiments identified important behavior and flow properties for shaken granular materials. Shaking was carried out both vertically and horizontally. The vertical component of shaking provided a means to modify gravity, and the horizontal component provided oscillating shear stress. This technique provides an excellent method to explore concepts of granular friction and failure. A detailed account of this work has appeared in Physical Review E, and in Europhysics Letters. More recently the task has focused on the properties of slowly sheared dense granular materials. Using a novel apparatus that allows characterization of all aspects of this process in 2 dimensions, we have discovered a novel transition that likely plays a key role in the failure of granular materials under applied stress. An initial account of this work has appeared in Physical Review Letters, and a more detailed account has appeared in Springer Lecture Notes in Physics and in a manuscript that is in preparation. This is a very complex system, and additional work is now being		

Task Description:	focused on understanding its properties. These explorations have recently extended to a much larger 2D system. The task was extend to include a flight mission using a 3D system. In this regard, a key point is that on earth gravity masks the true dynamical properties of both rapidly and slowly sheared granular systems. At this stage, however, the flight aspect of this project is no longer active, due to changes in NASA goals and missions. An additional key task, intimately related to the flight experiment, is a set of Molecular Dynamics (MD) studies in both 2D and 3D. These studies are of considerable scientific importance in their own right, and provide key insights into what might be expected in a microgravity environment. The task has been broadened over the past year by the addition of two new related projects. one of these is a 3D granular Couette experiment that is effectively a prototype for a flight experiment. The other is an experiment to probe the interaction of granular materials with a frictional boundary. This project has also focused on characterizing fluctuations and diffusion in 2D sheared systems, both experimentally and by MD simulation. Details of these results have recently appeared in Physical Review Letters, Physical Review E, European Physical Journal E, and Europhysics Letters. One final aspect has been the characterization of meteor-like impacts through the novel use of photoelastic materails. During the past several months, this project has lead to the discovery of a novel kind of 'phase'transiton between an ordered 3D crystal, and a dispordered phase. This transition, is seen in the shear apparatus, built as a progenitor to a flight experiment, that includes a mechanism to thermalize the system by shaking, as well as a shearing mechanism. The transition occurs because of competition between shaking, which, contrary to expectations, causes ordering, and shearing, which causes disordering. This transition is intimately connected to jamming and unjamming. Here, a particularly surprising discovery is the occurance of ordering when the system is shaken, hence, to certain extent, thermalized. This work has appeared in Physical Review Letters, and was presented at Powders and Grains 2005, in Stuttgart. It was selected for a special edition of Powders and Grains papers, and attracted considerable attention in the popular science press. This project entails the study of the properties of granular materials. The presence of gravity plays a strong role in how granular materials behave. In some instances, it also obscures fundamental behavior that could be much better understood in the presence of reduced gravity. The objectives of this work are to 1) understand the role that gravity plays in certain granular systems, such as sheared and shaken systems, and 2) to uncover basic science that governs the way granular materials behave and that is obscured ordinarily by the presence of gravity, 3) from the increased understanding from 1 and 2, increase our ability to predict and control granular flows, thus leading to significant savings in technical and commercial processing industries in the US and abroad, and 4) provide insight into granular materials handling in reduced-gravity situations. 5) In recognition of the initiative to explore the Moon and Mars, we are particularly focusing on a) improving our understanding of predictiability on granular materials, b) understanding the scope of granular processing for Lunar and Maritan exploration, and c) investigating novel systems of 2-phase granualar-gas flow that may be highly important for granular handling in the above environments. The science objectives have remained unchanged. The current key focus of this project is on the properties of sheared granular materials. Here, the key significance of the outcome of this task will be improvements in our understanding of the fundamental dynamical properties of granular materials. It is important to appreciate that our ability to describe these materials is primitive relative to what we can do for other materials such as Newtonian liquids. The negative consequences of this state of ignorance are substantial, and involve large financial and human losses. With a new focus on granular materials in Lunar and Martian settings, we expect to provide key insights that will be essential for the success of such missions. The shaking and shearing of granular materials are particularly important as basic means to cause flow in granular materials, and hence as tools to transport and process a variety of materials, including coal, ores, food grains, pharmaceutical powders and even dog food. These processes also plays key roles in natural phenomena such as avalanches and earthquakes. Gravity plays an intrinsically important role in these cases. Looking into the future, attempts to mine on the lunar surface will almost surely involve shaking processes for granular materials. Because our understanding of granular flows is rather poor, many of the plants and devices which handle granular materials operate significantly below capacity, and fail all too frequently. The result is corresponding loss to the US economy. The expectation is that increased scientific insight obtained from this project will lead to improved predictability and efficiency in a range of technological processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A deeper understanding of the properties of granular materials is essential for a broad spectrumof earth-bound tasks. Failures of devices on the Earth are all too common, and much insight is needed. The present tasks should help significantly towards this end.
Task Progress:	Per the discusion above, the project has yielded detailed information on the jamming-unjamming transition. Specifically, we understand this transition in terms of ordering, which occurs due to the extra energetic input supplied to the sample by shaking, and disordering/unjamming that occurs due to shearing. This process is also intimatley associated with dilation/contraction.
Bibliography Type:	Description: (Last Updated: 03/05/2024)
Abstracts for Journals and Proceedings	Daniels KE, Behringer RP. "Characterization of a freezing/melting transtion in a vibrated and sheared granular medium." Powders and Grains 2005, Stuttgart, Germany, July 18-22, 2005. Proceedings, Powders and Grains 2005, abstract 357. http://www.icp.uni-stuttgart.de/~pg2005/abstracts/Karen.html , Jul-2005
Abstracts for Journals and Proceedings	Utter BC, Behringer RP. "Diffusion, transients and non-affine deformations for granular Couette shear flow." Powders and Grains 2005, Stuttgart, Germany, July 18-22, 2005. Proceedings, Powders and Grains 2005, p. 197. http://www.icp.uni-stuttgart.de/~pg2005/abstracts/Utter.html , Jul-2005

Articles in Peer-reviewed Journals	Daniels KE, Behringer RP. "Hysteresis and competition between disorder and crystallization in sheared and vibrated granular flow." Physical Review Letters 2005 Apr;94(16):168001. http://dx.doi.org/10.1103/PhysRevLett.94.168001 , Apr-2005
Articles in Peer-reviewed Journals	Daniels KE, Behringer RP. "Characterization of a freezing/melting transition in a vibrated and sheared granular medium." Journal of Statistical Mechanics, in press March 2006. , Mar-2006